

DATA QUALITY AUDIT

Task 24 • Superstore Dataset

Professional audit report prepared with Python and Pandas

Rows Audited	9,977
Columns	13
Rules Executed	13
Rules Passed	12
Actionable Issues	1
Records Standardized	449

1. Executive Summary

The supplied dataset contains 9,977 records across 13 columns. The audit found no missing values, no full-row duplicates, no negative sales, invalid quantities, invalid discounts, unexpected categorical values, invalid category/sub-category combinations, or state/region mismatches.

Primary finding: 449 Postal Code values were represented as four-digit integers because leading zeros were not preserved. These values were standardized to five-character text in the cleaned sample.

Business note: Negative Profit is retained because it represents a legitimate loss-making transaction rather than a data-quality error.

2. Validation Results

Rule	Field	Failed	Status	Priority
DQ-01	All columns	0	PASS	—
DQ-02	All columns	0	PASS	—
DQ-03	Sales	0	PASS	—
DQ-04	Quantity	0	PASS	—
DQ-05	Discount	0	PASS	—
DQ-06	Postal Code	449	FAIL	Medium
DQ-07	Ship Mode	0	PASS	—
DQ-08	Segment	0	PASS	—
DQ-09	Country	0	PASS	—
DQ-10	Region	0	PASS	—
DQ-11	Category	0	PASS	—
DQ-12	Category / Sub-Category	0	PASS	—
DQ-13	State / Region	0	PASS	—

3. Issue Log

Issue	Field	Affected	Severity	Resolution
ISS-001	Postal Code	449	Medium	Postal Code stored as an integer instead of text.

4. Repeatable Quality Checklist

Profile the dataset → validate completeness and uniqueness → test numeric ranges → validate categories and cross-field relationships → standardize formats → document issues → correct → rerun all checks. This makes the audit repeatable rather than a one-time manual inspection.

Step	Control
1	Schema and data types
2	Missing/null values
3	Duplicate records
4	Numeric ranges and business rules
5	Categorical values

6	Cross-field consistency
7	Format and standardization
8	Correct, document, and retest

5. Deliverables

Excel audit workbook • PDF audit report • validation checklist CSV • issue log CSV • cleaned sample CSV • reusable Pandas audit script • GitHub-ready README and methodology notes.